# Tutorial: Potential energy surface exploration

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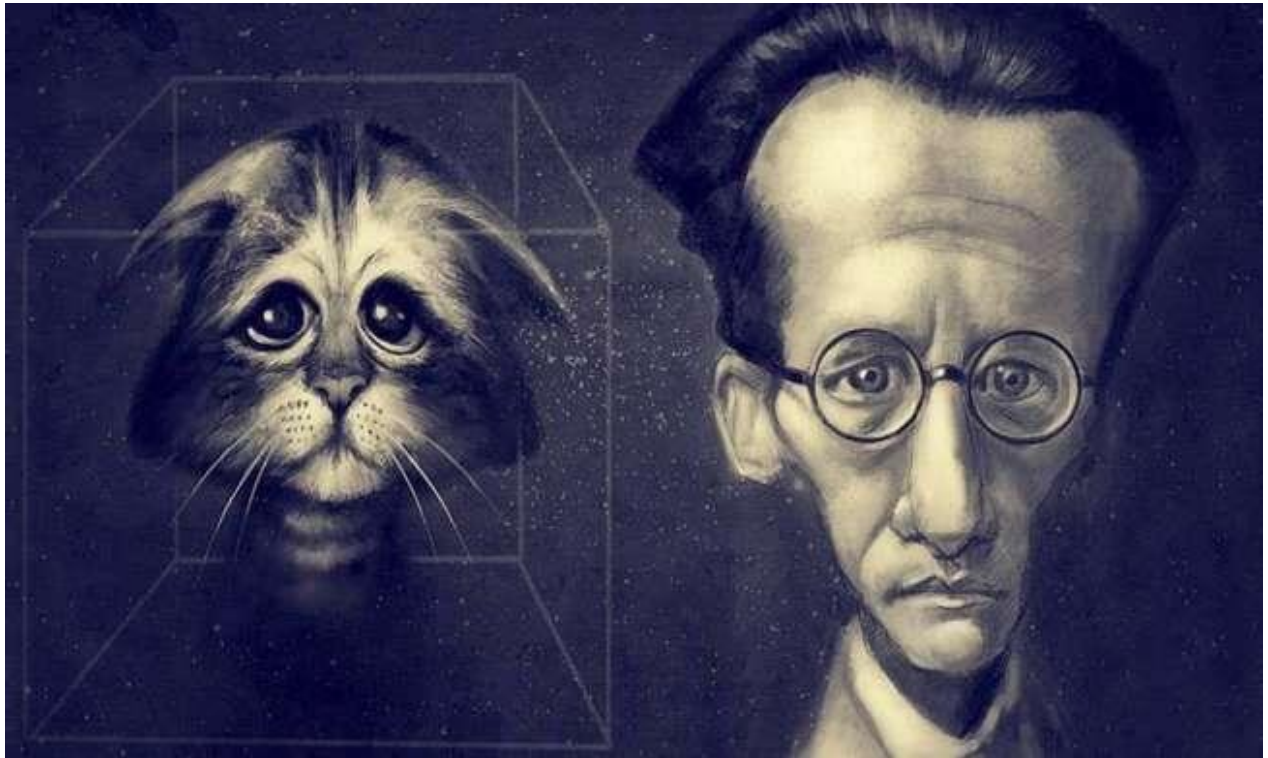
MOLAS

Molecular Level Atmospheric Science

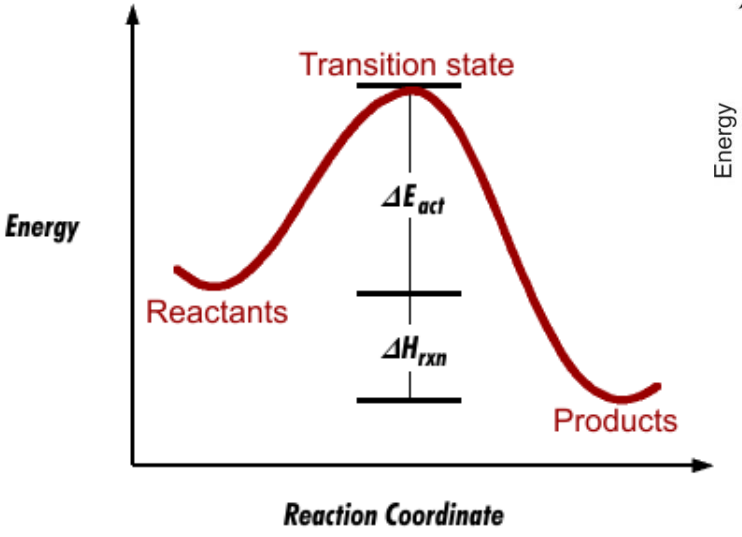
# Potential Energy Surface

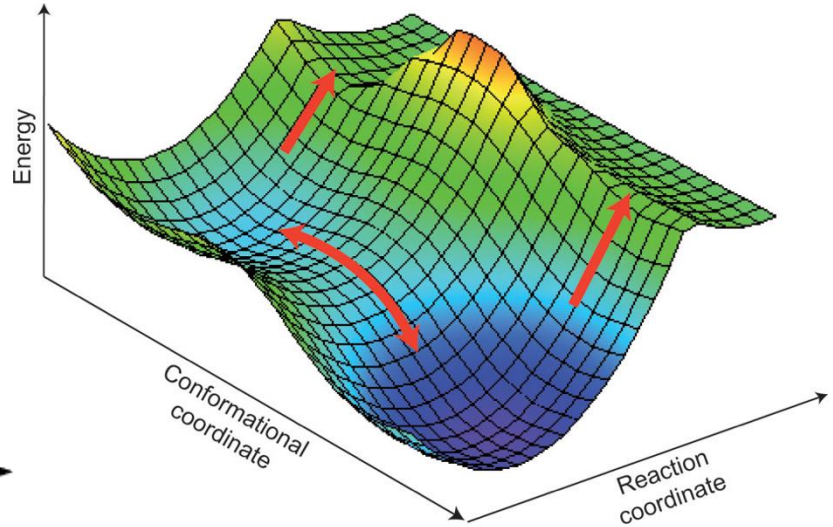
- Quantum chemistry is an important tool in studying fast, multi-step reactions
- Chemistry is the movement of electrons around atomic nuclei
  - ➢ Electronic structure calculations provide information about fundamental chemical properties



# Potential Energy Surface

- There are countless different approximations $\hat{\mathcal{H}}\Psi = E\Psi$
- *Born–Oppenheimer approximation:* The Schrödinger equation is solved separately for electrons, keeping the atomic nuclei in place
  - Each position of the atoms corresponds to a certain energy → potential energy surface




*Oppenheimer ja "Oppenheimer"*

# Molecular configurations

To perform meaningful calculations for molecules, usually the global minimum energy should be found.

*For small molecules, usually chemical intuition does the work. But for larger systems some configurational sampling methods are needed.*

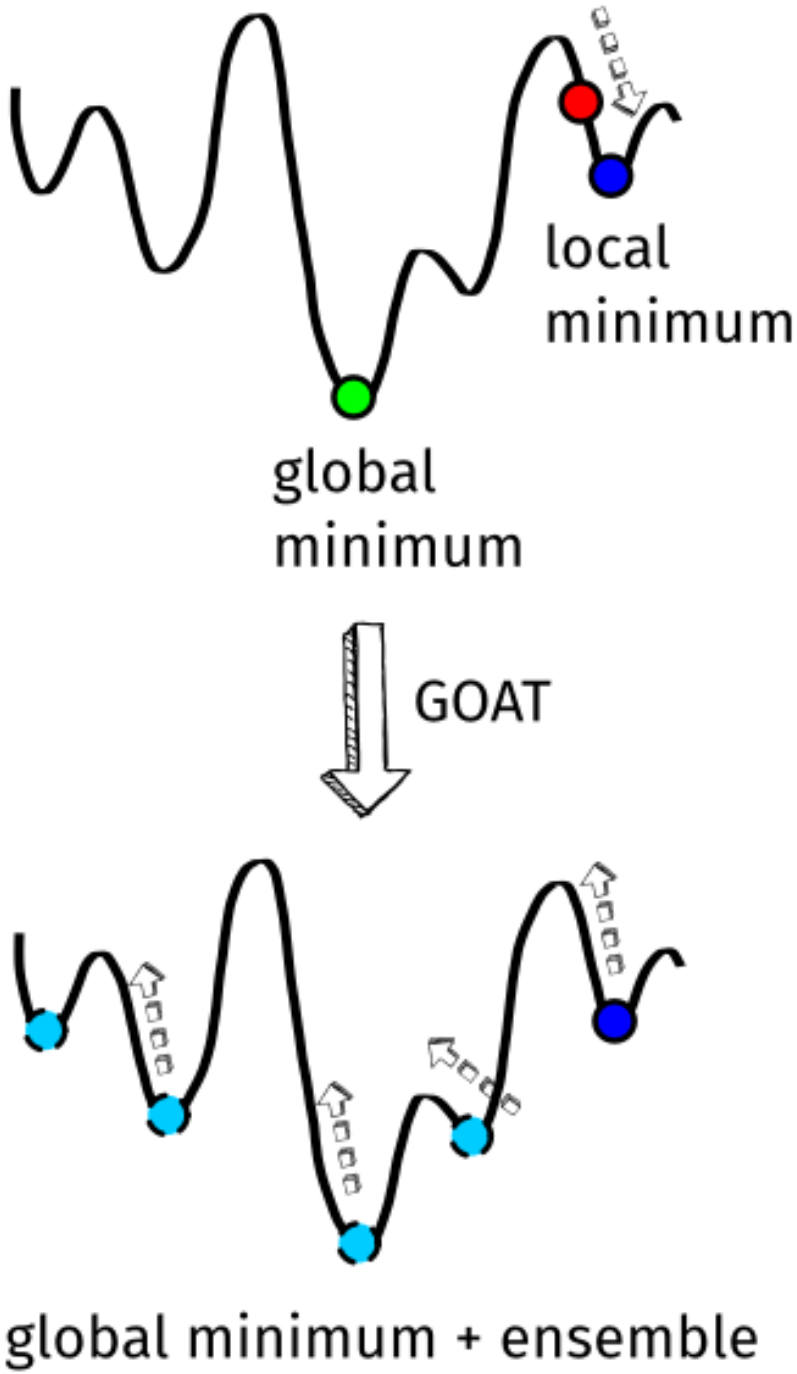
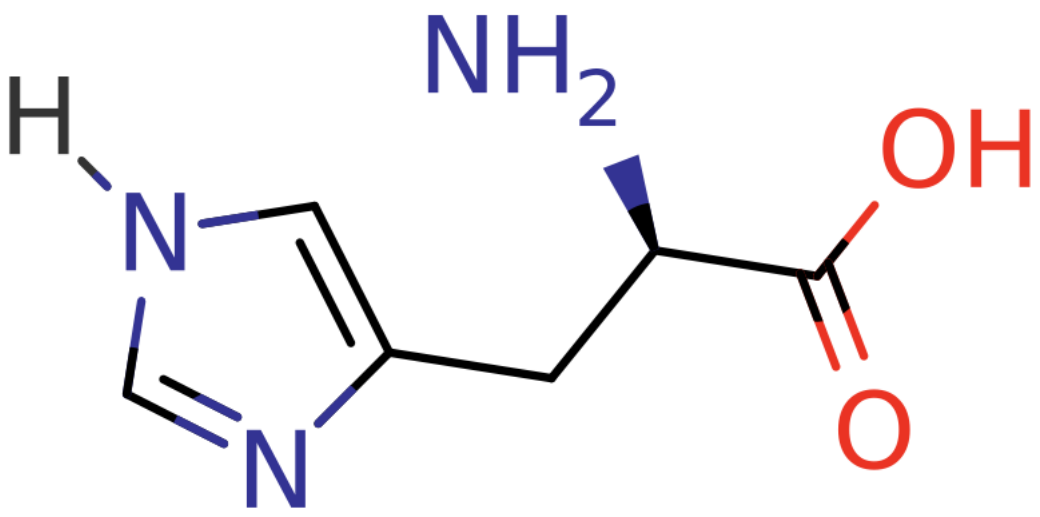



Fig: Orca 6 manual

# Molecular configurations

- Example: conformers of amino acid histidine



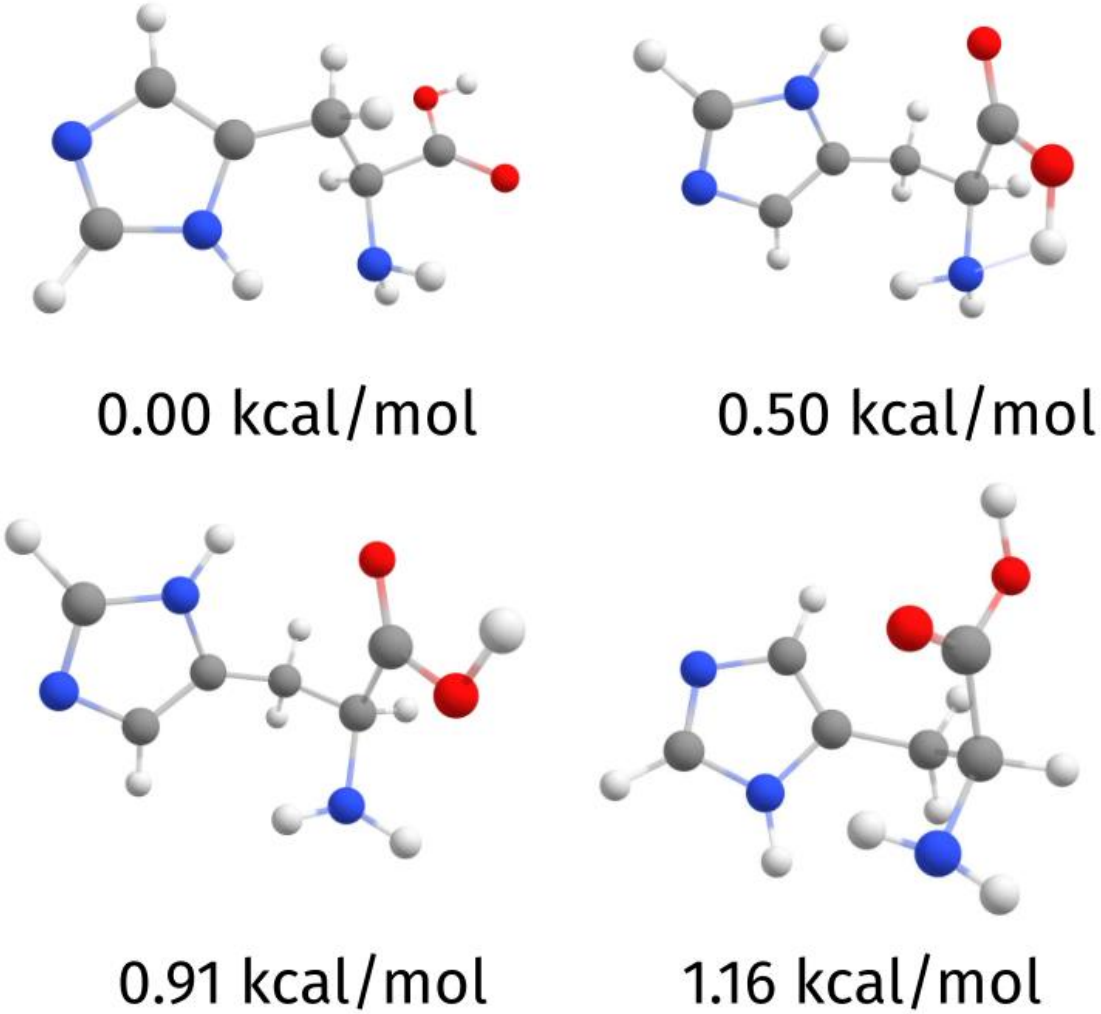

By simply looking at its Lewis structure, it is not at all evident that there are actually at least 20 conformers in the 3 kcal/mol range from the global minimum on the XTB PES!

# Molecular configurations

- Using high energy conformation to calculate e.g. reaction rate or FTIR spectrum might lead large errors and even wrong conclusions.

- Difference of 1 kcal/mol leads 10 times difference in reaction rates!
- Free and H-bonded OH stretches look different in spectrum!

*These are gas-phase minima; how you think situation change in aqueous phase?*




Four lowest energy conformers of histidine

# Ways to describe molecular flexibility

- *Conformer–rotamer ensemble sampling tool (CREST)*
  - Program for conformational sampling at the extended tight-binding XTB method
- Global optimizer algorithm (GOAT implemented in orca)
  - Can be used in principle with any method, in practice it is limited by computational cost (thus XTB recommended)
- Systematic flexibility based on molecular mechanics (e.g. in Spartan)
  - All possible rotations around single bonds: energy calculations based on force field

# How to find a global minimum structure?

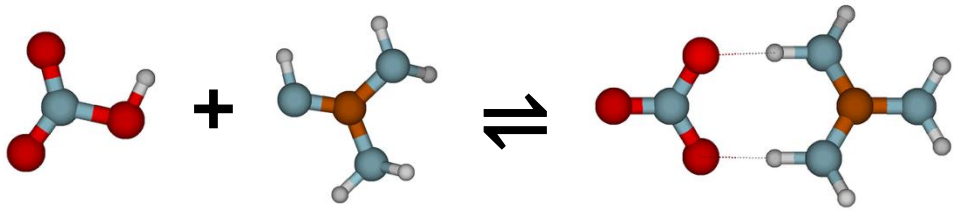
**Typical workflow:**

- Build a molecular structure
- Perform configurational sampling at low level of theory (fast calculations)
- Select several lowest energy structures and optimize them at medium level of theory
- Select a few lowest energy structures and perform final optimization and frequency calculation using desired accuracy
- Find the lowest (Gibbs free) energy structure (at certain temperature)
  - Accuracy can be increased even further by computing single point energy correction at high level of theory!

# Molecular interactions

- Interaction between molecules A and B define complex AB formation

  - For the complex formation reaction



$$\mathbf{A + B \rightleftharpoons AB}$$

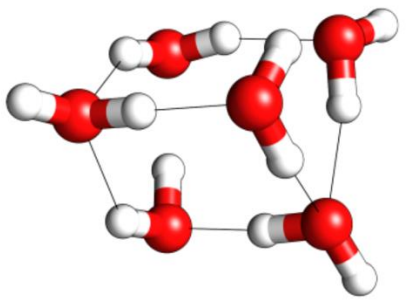
we can assume thermodynamic equilibrium $K = \frac{[\mathbf{AB}]}{[\mathbf{A}][\mathbf{B}]}$

and equilibrium concentration of an AB complex can be calculated as

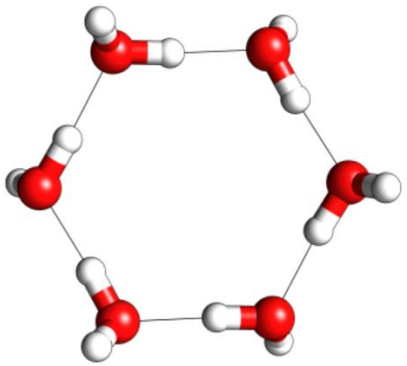
$$[\mathbf{AB}] = \frac{[\mathbf{A}][\mathbf{B}]}{[\mathbf{air}]} \exp\left(-\frac{\Delta G}{RT}\right)$$

The stronger the molecular interaction (more negative $\Delta G$), the higher the complex equilibrium concentration!

# How to compute interaction energy?

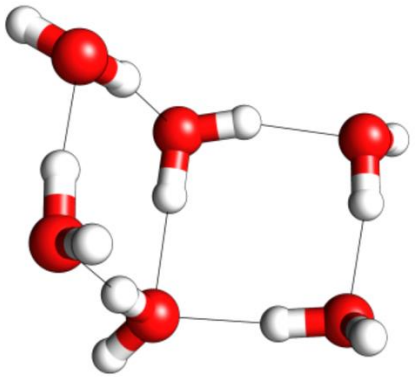


Computational chemistry predicts interaction energy by calculating energies of isolated monomers and the complex



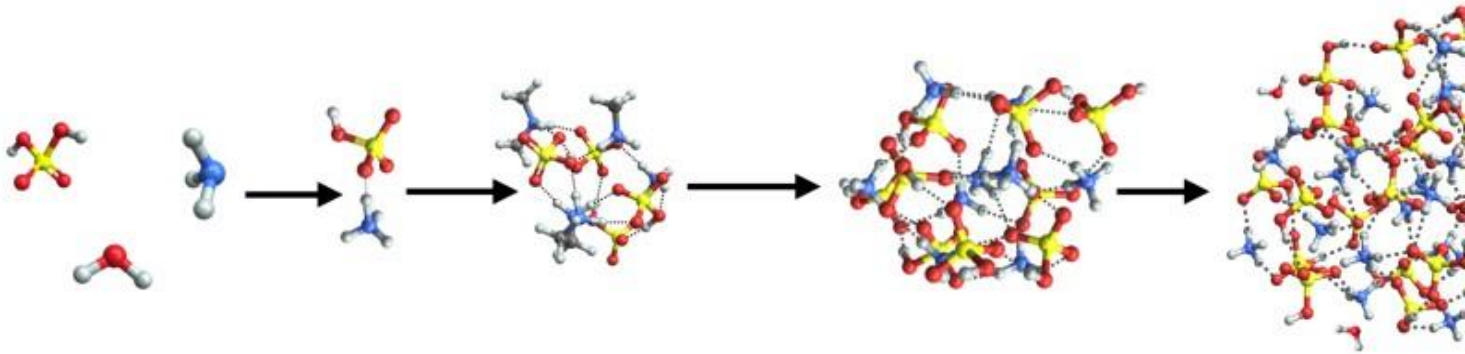
Typical workflow:

- Sample a set of initial complex structures using fast method
- Perform medium level optimization and select lowest energy structures
- Optimize geometries and calculate frequencies at desired accuracy
  - If needed higher accuracy, single point energy correction
  - Note: monomers should be calculated at the same level of theory!!!
- Calculate the difference between (the lowest energy) complex and its monomer energies



# Molecular interactions

- Gibbs free energy of formation is a key parameter
  - $\Delta G = \Delta H - T\Delta S$
    - $\Delta H$ ~ strength of intermolecular interactions (ion–ion, H-bonding, VdW)
    - $\Delta S$ ~ how much system loses flexibility (2 bodies become 1, ring structures)
- Not only molecular structures but also cluster structures matter
  - Needs to find global minimum structures to define energy accurately
  - Potential energy surface becomes highly complicated for large clusters

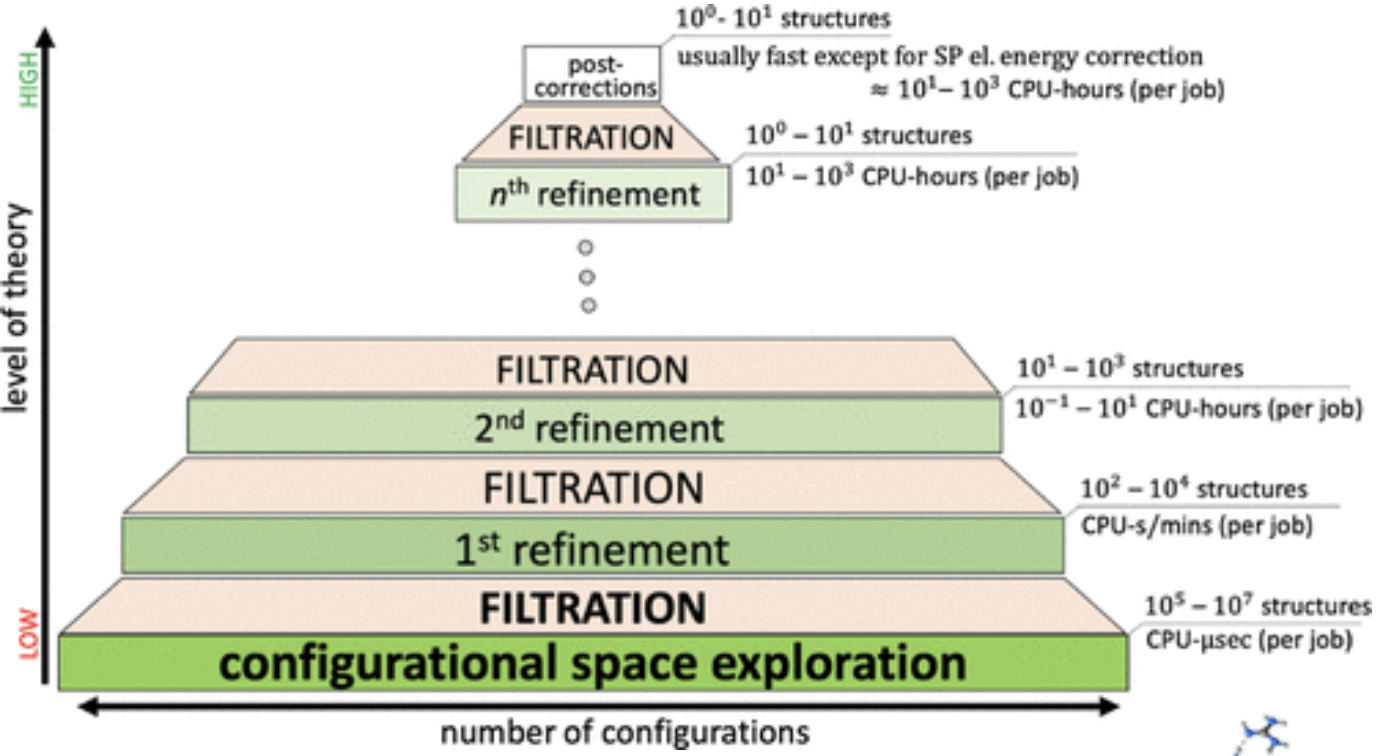


# Example procedure to sample large clusters

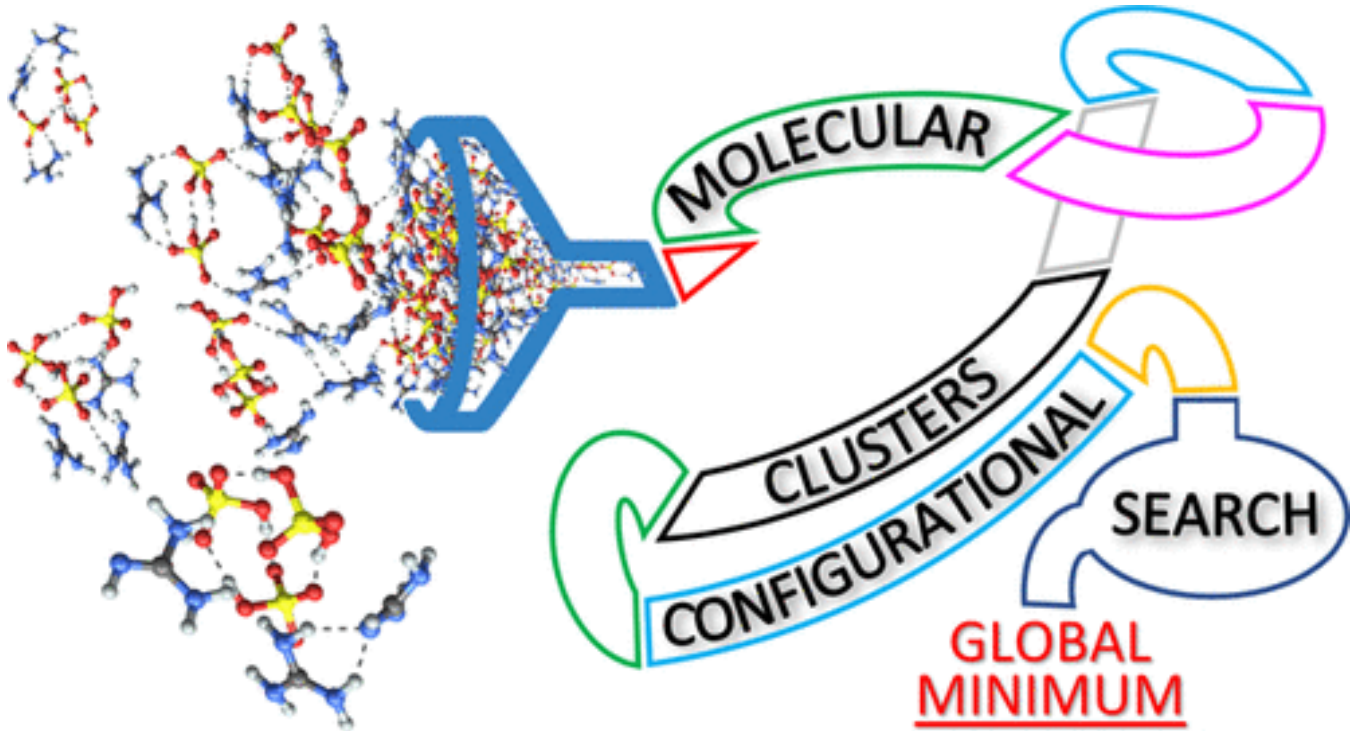
- Create 10000 initial structures with force field-based method
- Optimize all with rapid semi-empirical method
  - Filtering: select 1000 lowest energy unique conformers
- Calculate SPE with composite method
  - Filtering: select 100 lowest energy structures
- Pre-optimize with DFT
  - Filtering: select 10 lowest energy unique conformers
- Full optimization and frequency calculation with DFT
  - Filtering: select 1 lowest free energy conformer
- Calculate accurate SPE correction with coupled cluster method

JKCS

Kubečka et al., JPCA 2019




Kubečka et al., JPCA 2019



# Molecular clusters!

By Jakub Kubečka, Ivo Neefjes, and Vitus Besel from the University of Helsinki

https://www.youtube.com/watch?v=Kdrh82RVl3M&list=RDKdrh82RVl3M&start_radio=1

# Equilibrium constant

- For a cluster to be stable against evaporation at given conditions, it is required that the collision rate is equal to or higher than the evaporation rate
- According to the law of mass balance, the equilibrium constant K for reaction

$$\mathbf{A + B \rightleftharpoons C} \quad \text{is} \quad K = \frac{[C]^{\mathrm{eq}}}{[A]^{\mathrm{eq}}[B]^{\mathrm{eq}}} = \frac{k_{\mathrm{B}}T}{p_{\mathrm{ref}}} \exp\left(-\frac{\Delta G}{k_{\mathrm{B}}T}\right)$$

# Equilibrium constant

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at $T$=298 K and $p$=1 atm -> $2*10^{19}$ $cm^{-3}$

# Cluster evaporation and collision coefficients

- According to the law of mass balance, the equilibrium constant K for reaction

**A + B ⇌ C** is $$K = \frac{[C]^{\mathrm{eq}}}{[A]^{\mathrm{eq}}[B]^{\mathrm{eq}}} = \frac{k_{\mathrm{B}}T}{p_{\mathrm{ref}}}\exp\left(-\frac{\Delta G}{k_{\mathrm{B}}T}\right)$$

At equilibrium, the rate of cluster formation must be equal to cluster destruction:

$$\gamma_{(C\to A+B)}[C]^{\mathrm{eq}} = \beta_{A+B\to C}[A]^{\mathrm{eq}}[B]^{\mathrm{eq}}$$

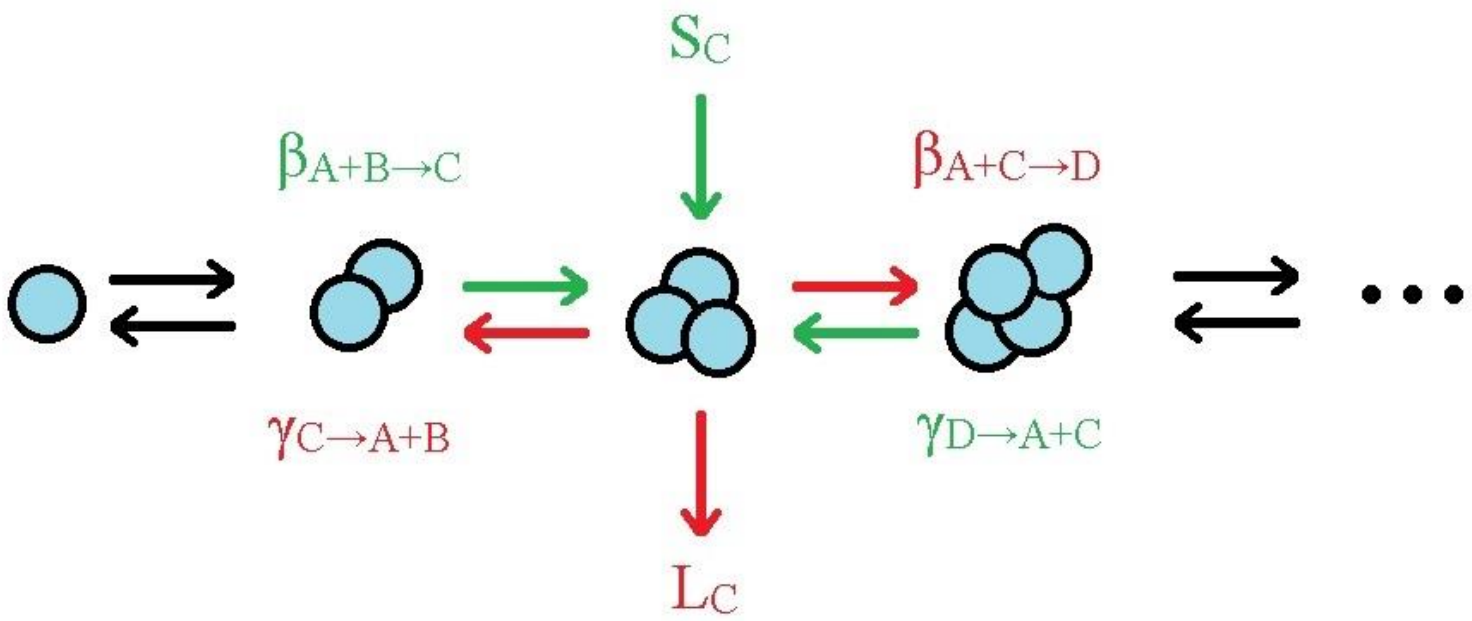
The evaporation rates of clusters can be obtained from Gibbs free energies:

$$\gamma_{(C\to A+B)} = \beta_{A+B\to C}\frac{p_{\mathrm{ref}}}{k_{\mathrm{B}}T}\exp\left(\frac{\Delta G}{k_{\mathrm{B}}T}\right)$$

Collision rates according to kinetic gas theory:

$$\beta_{A+B\to C} = \left(\frac{3}{4\pi}\right)^{1/6}\left[6k_{\mathrm{B}}T\left(\frac{1}{m_A}+\frac{1}{m_B}\right)\right]^{1/2}\left(V_A^{1/3}+V_B^{1/3}\right)^2$$

# What can be done with evaporation and collision coefficients?

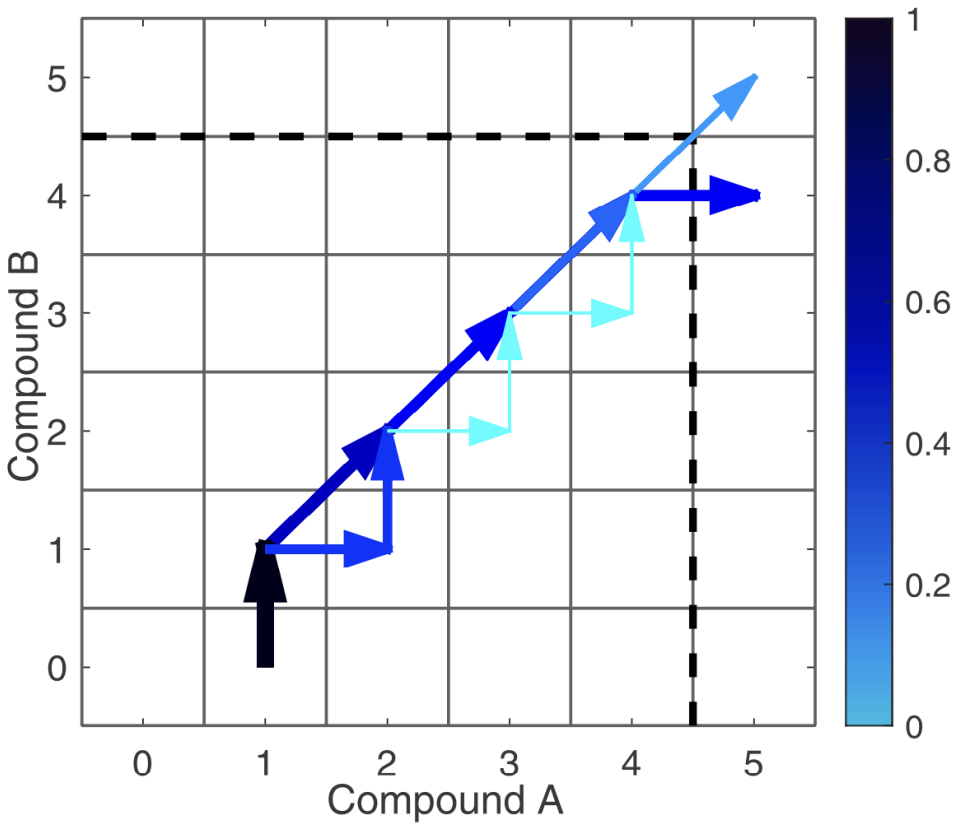



- Evaporation rates and collision rates are used to solve clusters' birth–death equations
- Includes all possible processes where the clusters can be formed (positive sign) or destroyed (negative sign)

**Birth–death equations**

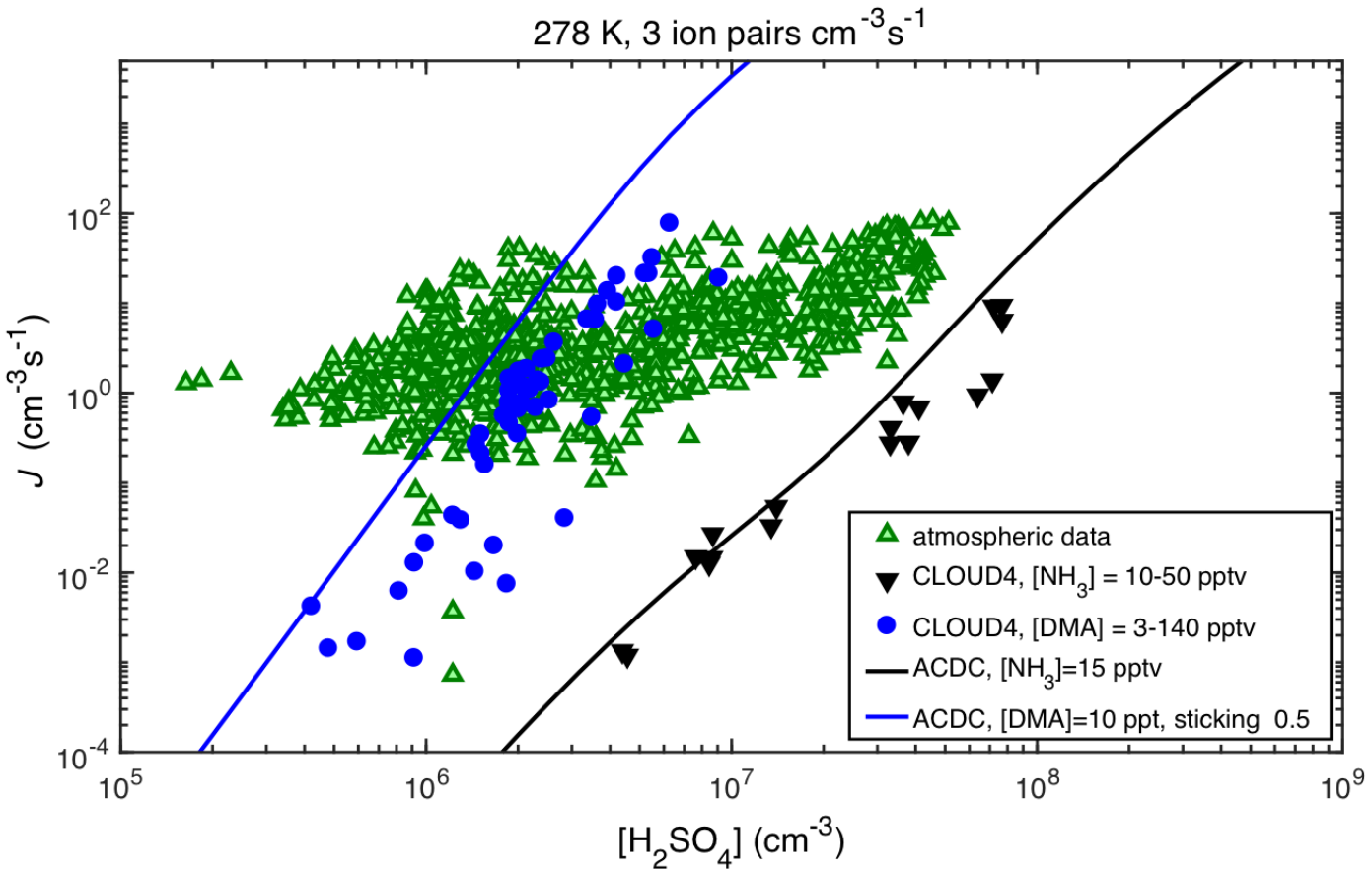
$$\frac{d\,[C]}{dt} = \sum_{C=A+B}^{A\leq B} \left(\beta_{A+B\to C}\,[A]\,[B] - \gamma_{C\to A+B}\,[C]\right) + \sum_{C=D-A}^{A\leq C} \left(\gamma_{(D)\to A+C}\,[D] - \beta_{A+C\to D}\,[A]\,[C]\right) + S_C - L_C\,[C]$$

# New-particle formation simulations



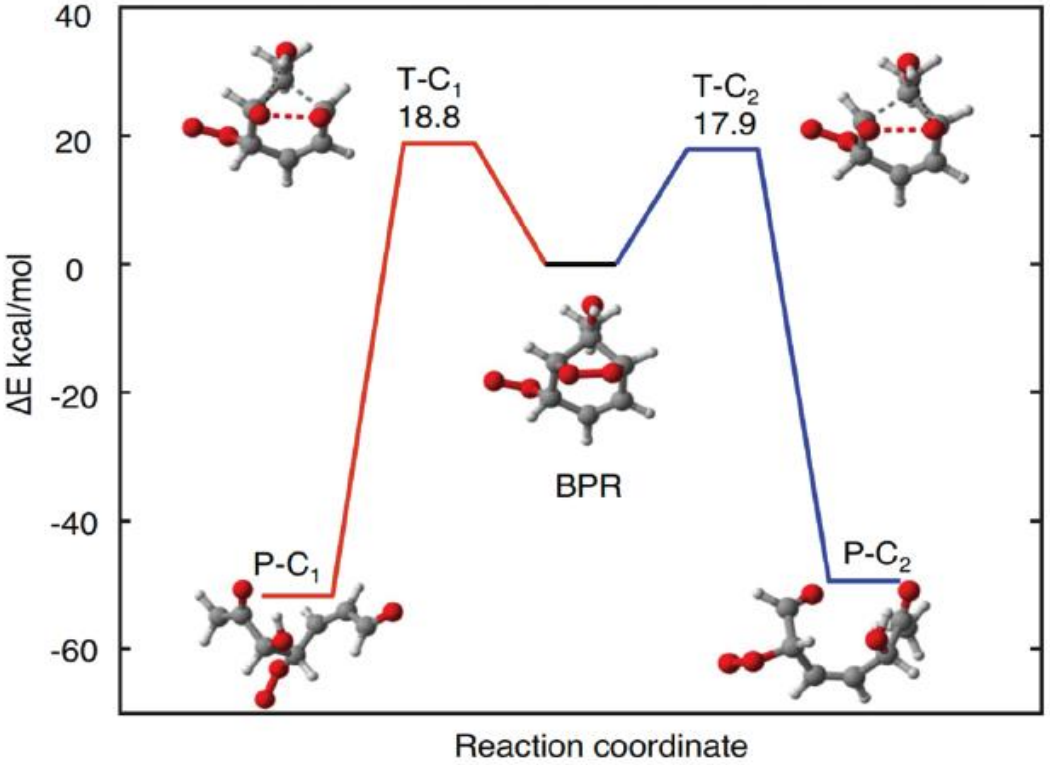

$$J = \frac{1}{2}\sum_{A}\sum_{B}\beta_{A,B}[A][B]_{\text{for } \{A+B \mid \text{condition}\}}$$

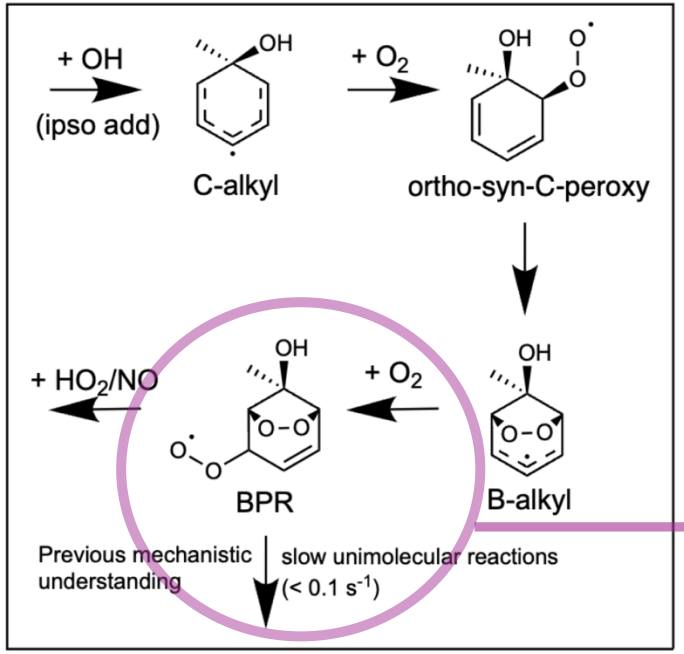
**Field vs lab vs simulated NPF rate**

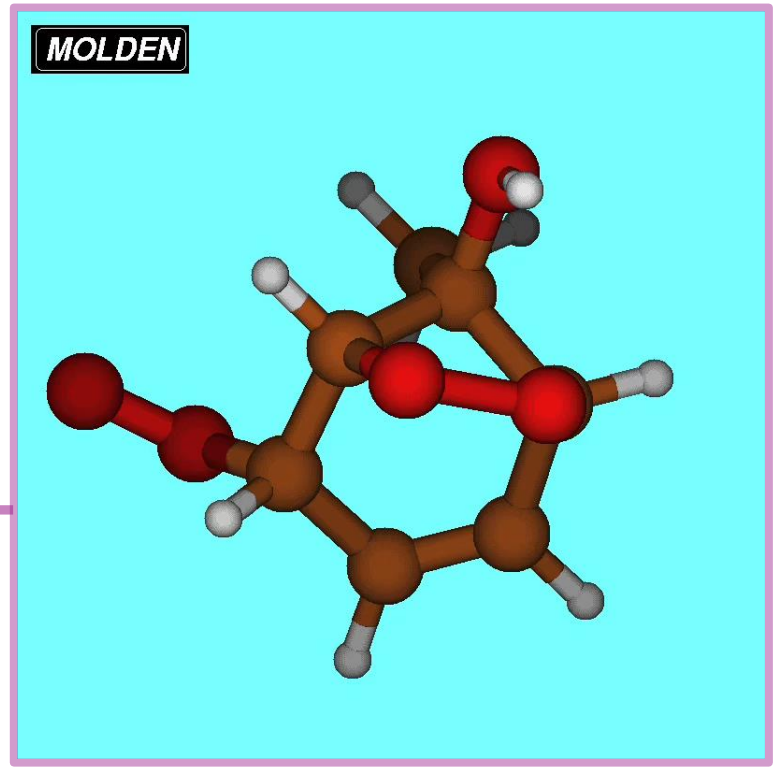


# Solving reaction mechanisms computationally

- Compute energy for reactants, transition states and products
  - Is reaction fast enough to happen?
  - Is reaction occurring spontaneously?



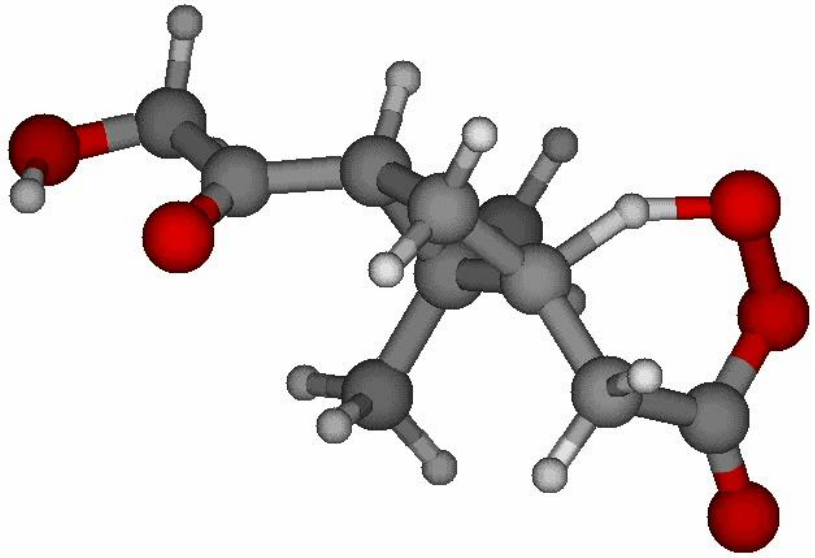




Iyer et al., Nat. Comm. 2023

# Example procedure to sample unimolecular transition states

- Create initial reactant structure and pre-optimize
- Find correct TS by scanning from reactant toward product
- Locate TS and perform TS optimization and frequency calculation
  - Check that imaginary frequency looks correct!
- Find other TS conformers with frozen CREST sampling
  - Filter out >3 kcal/mol energy conformers
- Loose frozen optimization with DFT
  - Filter out duplicates and >2 kcal/mol energy conformers
- Full TS optimization and frequency calculation
  - Select the lowest energy TS conformer
  - Perform IRC calculation
- Calculate SPE correction with coupled cluster method



Pasik et al. PCCP 2024

# Example procedure to calculate rate coefficient

- Perform conformer sampling also for the reactant
- Calculate the final energies for lowest energy conformers at the same level as for TS
- Rate coefficient can be now calculated using MC-TST
  - Gibbs free barrier energy directly from calculations
  - Tunneling factor $\kappa$ is needed for H-shift reactions

$$k(T) = \kappa \frac{k_\mathrm{B} T}{h} (c^{\ominus})^{1-M} e^{-\Delta G^{\ddagger}/RT}$$

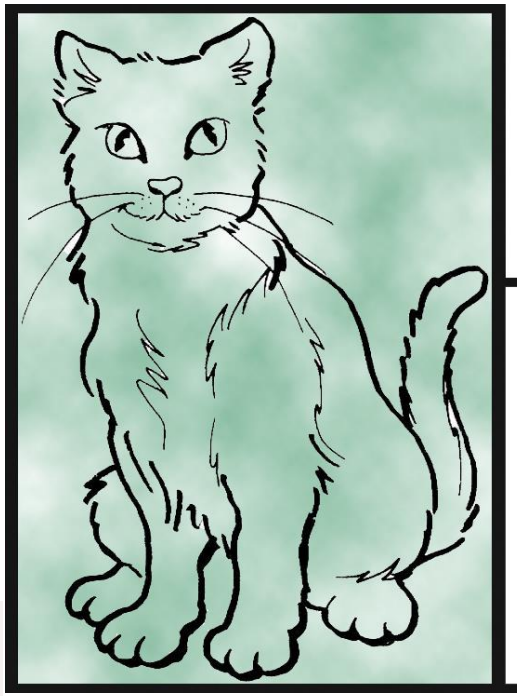
The factor $k_\mathrm{B}T/h$ gives the frequency of molecular collision

The factor $(c^{\ominus})^{1-M}$ ensures the dimensional correctness of the rate constant

# Schrödinger's cat

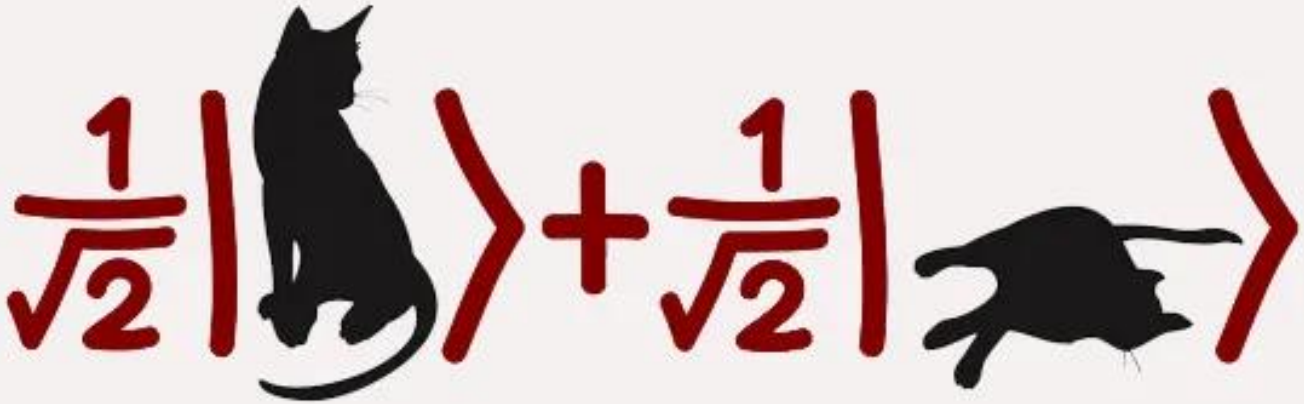
*– connection to Nino's favorite methods*

- Probability in quantum mechanics is written as a linear combination of all possible states



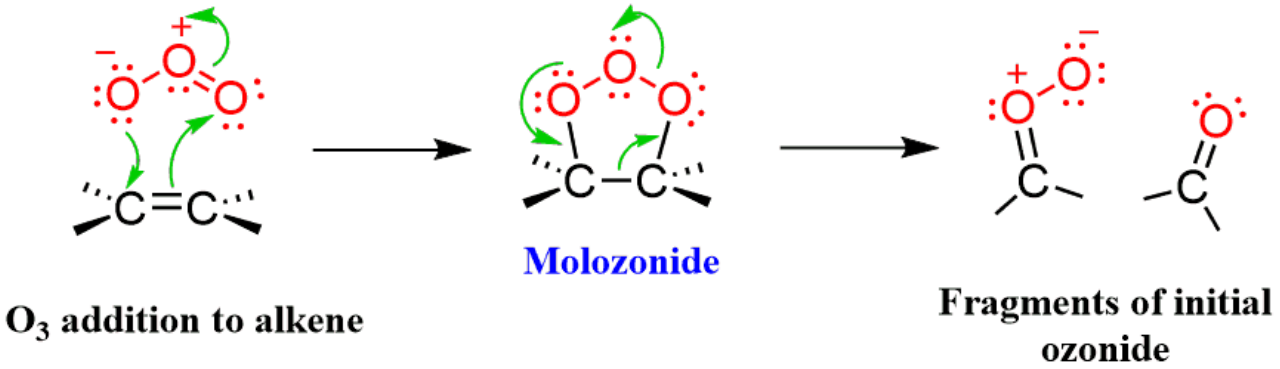
50% chance that the poison will be released within an hour and kill the cat.

**For cat:**

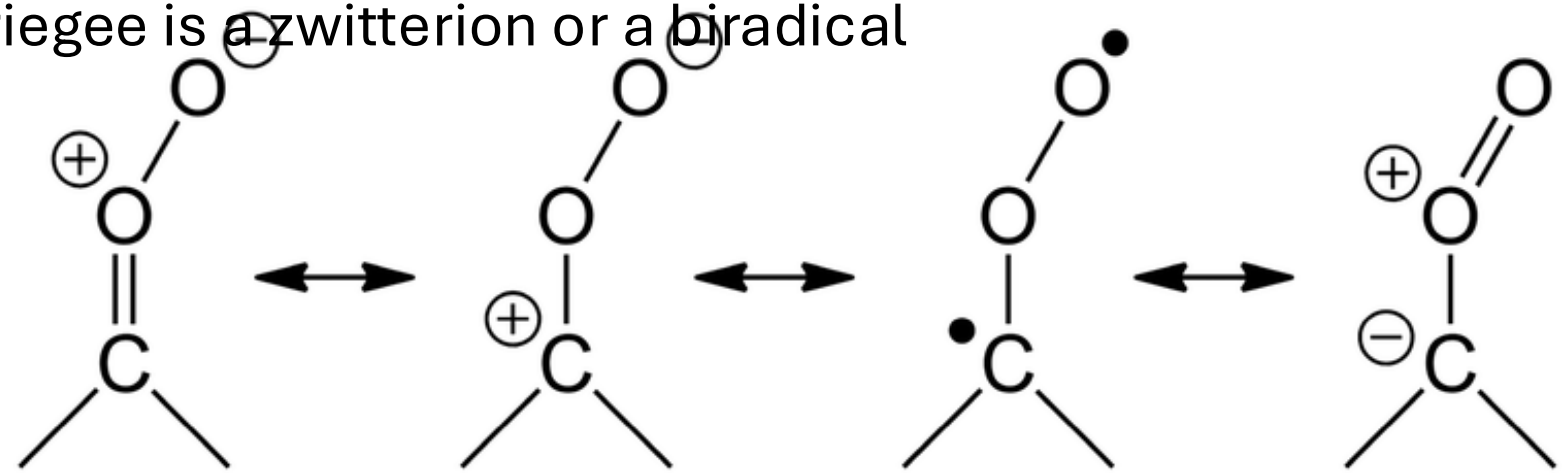


# Criegee intermediate

- Formed by the ozonation of an alkene
- Important intermediate in atmospheric chemistry reactions




- Electronic structure is complex: several resonance structures
  - Without "measurement" it is not possible to know whether Criegee is a zwitterion or a biradical



# Useful programs

SOFTWARE FOCUS | Open Access

**Software Update: The ORCA Program System—Version 6.0**

Frank Neese

First published: 28 April 2025 | https://doi.org/10.1002/wcms.70019 | VIEW METRICS

https://orca-manual.mpi-muelheim.mpg.de/#

**CREST—A program for the exploration of low-energy molecular chemical space**

**Special Collection:** Modular and Interoperable Software for Chemical Physics

Philipp Pracht; Stefan Grimme; Christoph Bannwarth; Fabian Bohle; Sebastian Ehlert; Gereon Feldmann; Johannes Gorges; Marcel Müller; Tim Neudecker; Christoph Plett; Sebastian Spicher; Pit Steinbach; Patryk A. Wesołowski; Felix Zeller

Check for updates

+ Author & Article Information

*J. Chem. Phys.* 160, 114110 (2024)

https://doi.org/10.1063/5.0197592 **Article history**

https://crest-lab.github.io/crest-docs/

Issue 37, 2015

From the journal:

Physical Chemistry Chemical Physics

**ABCluster: the artificial bee colony algorithm for cluster global optimization**

Jun Zhang *[a] and Michael Dolg *[a]

https://www.zhjun-sci.com/abcluster/doc/index.html

*The Journal of Physical Chemistry A* > Vol 123/Issue 28 > Article

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Jakub Kubečka*, Vitus Besel, Theo Kurtén, Nanna Myllys, and Hanna Vehkamäki

https://jkcs.readthedocs.io/en/latest/index.html

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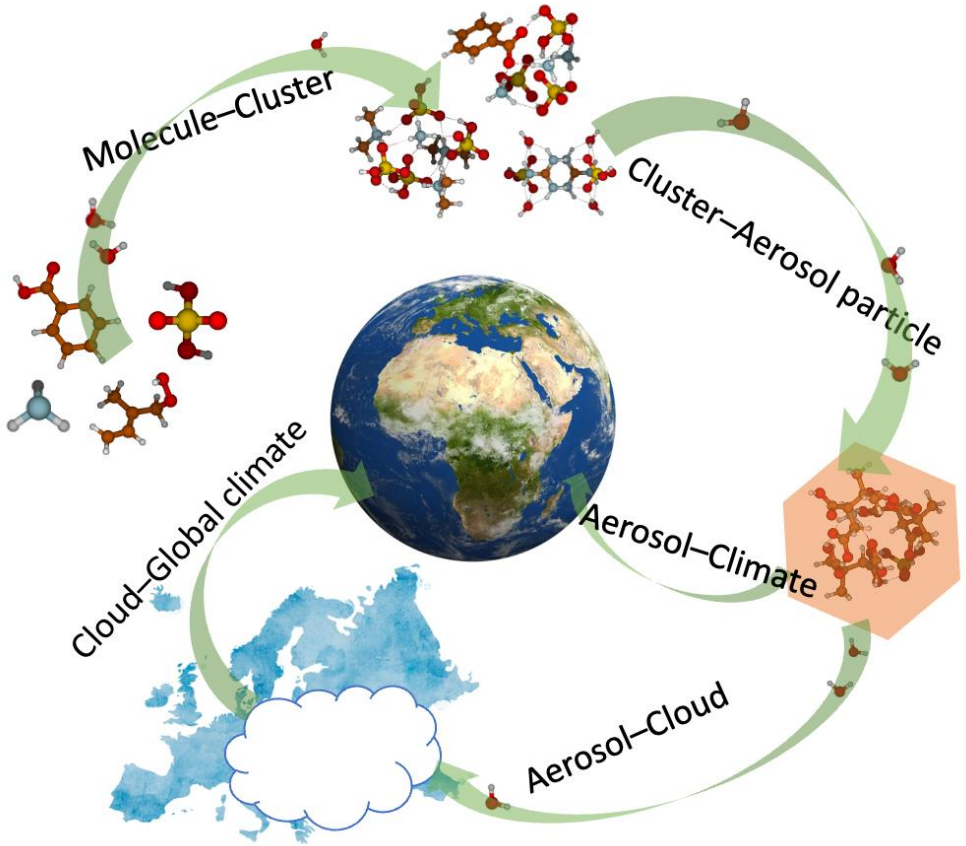


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